

DENVar: A Global Dynamic Estimate of NDVI Variance

Stefano Mezzini^{1,2}, Gavin L Simpson³, and Michael J Noonan^{1,2,4,*}

¹Okanagan Institute for Biodiversity, Resilience, and Ecosystem Services, The University of British Columbia Okanagan, Kelowna, British Columbia, Canada.

²Department of Biology, The University of British Columbia Okanagan, Kelowna, British Columbia, Canada.

³Department of Animal and Veterinary Sciences, Aarhus University, Denmark.

⁴Department of Computer Science, Math, Physics, and Statistics, The University of British Columbia Okanagan, Kelowna, British Columbia, Canada.

*Address correspondence to: michael.noonan@ubc.ca

Abstract

Remote-sensing proxies of habitat productivity such as the Normalized Difference Vegetation Index (NDVI) have helped develop and support many hypotheses and projects. However, historically, users tended to focus solely on trends in mean productivity. In recent years, the variance around the mean (i.e., stochasticity) has gained more attention, particularly as climate change and human-induced rapid environmental change transform the ecosystems species have evolved in and adapted to. Still, calculating trends in NDVI stochasticity requires careful modeling, since stochasticity estimates are conditional on the estimated mean and the scale at which the data are collected and modeled. We thus present a new global Dynamic Estimate of NDVI Variance, *DENVar*. We estimate spatiotemporal trends in NDVI mean and variance using Generalized Additive Models, which provide a flexible yet transparent and interpretable model structure. We show that *DENVar* varies nonlinearly with **XXX** and can be used to test many hypotheses related to environmental stochasticity, such as optimal foraging theory and regime shifts as well as for conservation. We provide a publicly and freely available repository for downloading rasters of daily and long-term averages of *DENVar* and the related estimates of mean NDVI.

1 Introduction

The Normalized Difference Vegetation Index (NDVI) is a proxy of primary productivity used in a wide range of disciplines, including ecology (Pettorelli et al., 2005, 2011), landscape phenology (Merkle et al., 2016; Hmimina et al., 2013), agriculture (Franch et al., 2017), forestry (João et al., 2018), and conservation (Mallegowda et al., 2015; Marcus et al., 2025). High-resolution, remotely-sensed NDVI data have allowed analysts to estimate plant biomass without fieldwork (Bhatti et al., 2024; Borowik et al., 2013) and leverage such estimates for studying large-scale plant dynamics (Pettorelli et al., 2005) and their effects on animal movement behavior (Merkle et al., 2016; Aikens et al., 2017). Still, most NDVI work is limited to quantifying changes in and responses to mean NDVI while ignoring changes in the variance around the mean.

Environmental stochasticity has been studied for decades (Levins, 1974; Stephens and Charnov, 1982; Lande, 1993; Sæther, 1997), but recent work has drawn more attention to the effects environmental stochasticity has at a wide range of scales, from individual organisms (Mezzini et al., 2025; Rizzuto et al., 2021) to landscapes (Zhang et al., 2009; Mutti et al., 2020; Marcus et al., 2025). However, estimating trends in NDVI variance requires more nuanced modeling than estimating changes in mean NDVI, particularly when environments are temporally dynamic and spatially heterogeneous. The scale at which processes are observed determine how individuals’ perception of and ability to respond to change (Levin, 1992). A forest fire is severely disruptive and deadly for many insects (Fettig et al., 2022), while birds and large mammal may not struggle to escape the fire and or establish a new range (Franklin et al., 2021; Calhoun et al., 2024). Similarly, the effects of the fire recovery will depend on its duration relative to other organisms’ phenology and lifespan (Frankham and Brook, 2004). Therefore, distinguishing between a change in mean NDVI and a stochastic fluctuation requires a conscious decision about the scale at which to consider the data.

Previous work suggests that environmental stochasticity can reduce limit species’ ability to adapt or specialize (Chevin et al., 2010), alter species’ movement behavior (Mezzini et al., 2025), and even affect extinction risk at both the population and species levels (Lande, 1993; Bull et al., 2007; Fung et al., 2018). Still, the effects of environmental stochasticity on species dynamics and ecosystem stability remain understudied (Marcus and Noonan, 2023). Global, temporally dynamic estimates of NDVI variance would facilitate research by building upon the widespread use and interpretability of NDVI while also removing the barriers of computational demand and statistical knowledge that are required for quantifying environmental stochasticity. However, since NDVI is not a direct measurement primary productivity but a proxy, it should be interpreted carefully (Pettorelli et al., 2011). NDVI is prone to sensitivity loss in very green environments (Matsushita et al., 2007; Son et al., 2014), values vary across sensors (Beck et al., 2011; Fan and Liu, 2016; Huang et al., 2021), and the coarse spatiotemporal scale of satellite-derived NDVI can be a substantial barrier when assessing fine-scale trends (Goodchild and Quattrochi, 2023; Reid et al., 2018). However, careful statistical modeling can help overcome some of these limitations. In particular, Generalized Additive Models (GAMs) are particularly convenient due to their ability to learn flexibly from data (Wood, 2017; Simpson, 2018) and handle very large spatiotemporal datasets (Wood et al., 2015, 2017).

We present a new measure of environmental stochasticity, which we name the Dynamic Estimate

of NDVI Variance (DENVar). We calculate DENVar using 44 years of NDVI data from the AVHRR and VIIRS collection of sensors and GAMs. Although satellites with AVHRR and VIIRS sensors provide data at a coarser spatial resolution (0.05°) than other satellites (e.g., 250m for MODIS), they are the only sensors to provide data at a daily frequency and for more than 40 years. Our use of GAMs provides a flexible yet interpretable approach for estimating trends in mean and variance in NDVI without the rigidity of other more traditional methods (e.g., ARIMA) or the opaqueness of machine learning models such as neural networks (Simpson, 2018; Wood, 2017). We illustrate the relationships between DENVar and multiple, relevant ecological metrics, and we provide access to a free repository of daily, global estimates of DENVar and the corresponding estimates of mean NDVI.

2 Methods

2.1 Choice of color palettes

We represent NDVI using a modified version of Fabio Crameri’s divergent *bukavu* palette (Crameri, 2018a; Crameri et al., 2020), which has high-contrast for deuteranope and protanope vision. For NDVI values between -0.1 and 1, the colors also have sufficient contrast for the colors to also be distinguishable by strong tritanope and achromatic vision. Fig. A.1 contains an approximate representation of how each vision type would perceive each color palettes used. We obtained all color palettes with three or more colors from the *khroma* package (v. 1.14.0 Frerebeau, 2024) for R (v. 4.4.0; R Core Team, 2024).

2.2 Datasets used

Fig. 1 illustrates the processes of data collection, cleaning, and modeling used in this project. We downloaded NDVI image composites directly from the servers for data from AVHRR sensors (1981-06-24 to 2013-12-31; United States National Centers for Environmental Information (NCEI) and United States National Aeronautics and Space Administration, 2025; Vermote and NOAA CDR Program, 2018) and VIIRS sensors (2014-01-01 to 2025-06-29; United States National Aeronautics and Space Administration, 2025; Vermote and NOAA CDR Program, 2022).

We removed all pixels labeled as "cloudy" for AVHRR rasters and "probably cloudy" or "confidently cloudy" for VIIRS rasters. To remove pixels that included large portions of water, we used a shapefile of major water bodies (Garmin International, Inc. and Esri, 2025) to produce a static raster of the percentage of water within each pixel, at the same spatial resolution as the NDVI rasters (0.05° ; Fig. A.2). We removed all pixels with $> 40\%$ water cover.

Many rasters had horizontal bands of $\text{NDVI} > 0.9$ at unrealistically northern latitudes between September and March, likely due to limited visibility during fall and winter in the Northern hemisphere. We identified and removed problematic pixels using the monthly empirical cumulative distribution functions of NDVI across latitude (Fig. A.3). We did not filter problematic values at latitudes $< 60^\circ\text{S}$ since we fully excluded Antarctica from the analyses (Fig. A.4).

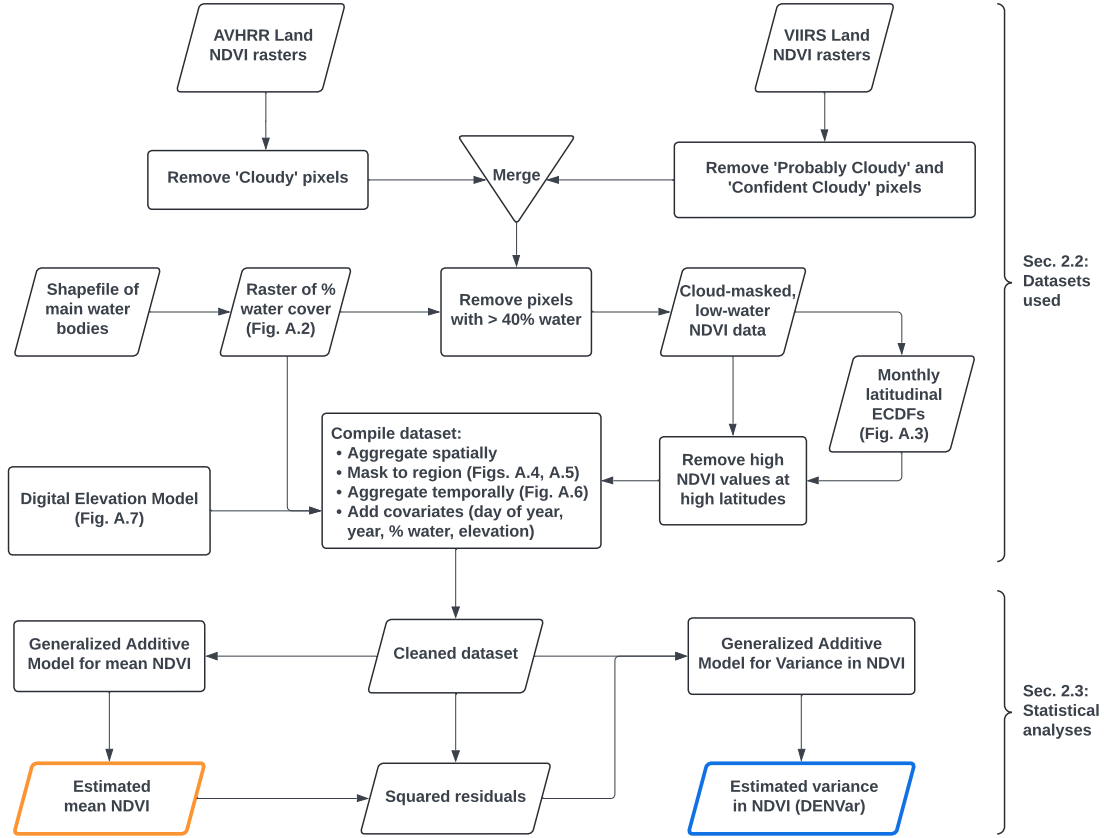


Figure 1: **Flowchart for the analyses used in the manuscript.** The curly braces on the right indicate the subsections in the Methods section that detail the dataset collation and cleaning (2.2), and statistical analyses (2.3).

To ensure the number of rows for each dataset was below the maximum data frame size in R ($3.17 \times 10^{10} > 2^{31} - 1 \approx 0.21 \times 10^{10}$), we reduced the dataset size by splitting it across four spatial groups (Figs. A.4 and A.5) and spatially and temporally aggregated the rasters by a factor of four. The final dataset thus had one raster every four days with a spatial resolution of $0.05^\circ \times 4 = 0.2^\circ$ (Fig. A.6). Finally, we annotated the dataset with columns of day of year (1-366), integer year (1981-2025), percent water cover, and elevation above sea level (m). The digital elevation model was obtained using the `get_elev_raster()` function from the `elevatr` package for R (v. 0.99.0; Hollister et al., 2023) with an original resolution of 0.076° (Fig. A.6). Despite the substantial reduction in dataset size, the dataset creation was still challenging for the machine used, which had 2.2 TB of RAM and two Intel Xeon Platinum 8462Y+ processors (64 cores total).

2.3 Statistical Analyses

We estimated spatiotemporal trends in mean and variance in NDVI using Generalized Additive Models (GAMs) via the `mgcv` package for R (v. 1.9-3 Wood, 2017). To reduce modeling fitting times

with negligible losses to model accuracy, we used the `bam()` function with fast REstricted Marginal Likelihood (`method = fREML`) and covariate discretization (`discrete = TRUE`; Wood et al., 2015, 2017). Ideally, NDVI should be modeled using beta location-scale models (after the linear transformation $Y^* = (Y + 1)/2$) to account for the fact that: (1) NDVI is bounded between -1 and 1, and (2) the variance in NDVI is dependent on the mean (and vice-versa), since ecosystems with very low NDVI (e.g., rock, ice, or concrete) or very high NDVI (e.g., dense forest) tend to have lower variance in NDVI. However, preliminary tests and work by (Marcus et al., 2025) showed that beta GAMs and particularly beta location-scale GAMs had prohibitive requirements for both fitting times and memory requirements, since both requirements scale $O(Mnp^2)$, for M smoothing parameters and a model matrix with n rows and p columns (Wood et al., 2015). In contrast, Gaussian models with fixed scaled parameters (i.e., constant variance) fit substantially faster and provided very similar spatialtemporal estimates of mean and variance in NDVI (Appendix B). Thus, we estimated trends in mean and variance in NDVI using a pair of GAMs for each region shown in Fig. A.4; one for estimating mean NDVI over space and time (assuming a constant variance), and one for estimating the mean squared residual (i.e., variance) in NDVI (again, assuming a constant scale parameter).

Together, the sets of models for mean NDVI (i.e., μ) and variance in NDVI (i.e., σ^2) had the following structure, with ν indicating NDVI:

$$\left\{ \begin{array}{l} \nu \sim \text{Normal}(\mu, \sigma^2) \\ \mu = \mathbb{E}(\nu) \approx \hat{\mathbb{E}}(\nu) = \hat{\mu} \\ \hat{\mu} = \beta_{\mu} + s(\text{prop_water}) + s(\text{elev_m}) + s(\text{doy}) + s(\text{year}) + s(\text{lat, long}) + \\ \quad t(\text{doy, year}) + t(\text{doy, lat, long}) + t(\text{year, lat, long}) \\ \sigma^2 = \mathbb{E}((\nu - \mu)^2) \approx \hat{\mathbb{E}}((\nu - \hat{\mu})^2) = \hat{\sigma}^2 \\ \hat{\sigma}^2 = \beta_{\sigma^2} + s(\text{prop_water}) + s(\text{elev_m}) + s(\text{doy}) + s(\text{year}) + s(\text{lat, long}) + \\ \quad t(\text{doy, year}) + t(\text{doy, lat, long}) + t(\text{year, lat, long}) + s(\hat{\mu}) \end{array} \right., \quad (1)$$

where, for a given (spatiotemporally aggregated) pixel, `prop_water` is the proportion of water within the pixel, `elev_m` is elevation above sea level in meters, `doy` is integer day of year (1-366), `year` is integer year (1981-2025), `lat` and `long` are latitude and longitude, respectively. For ease of readability, we omit the pixel-level indices. $s(\)$ indicates marginal smooth functions, while $t(\)$ indicates tensor product interaction terms. Seasonal terms used cyclical cubic splines (`bs = 'cc'`) with edge knots 0.5 and 366.5, while spatial smooths used a basis with second-order spline on the sphere (`bs = 'sos'`; Wood, 2017). The linear predictors for μ and σ^2 have the same structure, other than the $s(\mu)$ smooth term for σ^2 , but the functions for σ^2 are independent of those for μ . Despite the data aggregation procedures and faster fitting methods, including parallelizing computation across 60 cores, each of the four models required multiple days to fit. Predicting from the models for $\hat{\mu}$ was also had substnatial time and memory demands, since predictions were calculated without discrete prediction methods (`discrete = FALSE`). While discrete predictions were orders of magnitude faster,

145 they were excessively spatially coarse. See Appendix C and the code availability statement for
146 additional information.

147 Since there are no corrections to any bias in residuals from models 1, the estimates of variance
148 from models 1 may be inflated near areas or periods of rapid and substantial change. Still, the
149 appreciability of this inflation is dependent on the spatiotemporal scale over which changes are
150 perceived (Levin, 1992; Frankham and Brook, 2004; Goodchild and Quattrochi, 2023; Mezzini et al.,
151 2025).

152 From the two sets of models, we produced daily estimates of (1) the Dynamic Estimate of NDVI
153 Variance (DENVar) and (2) the estimated mean NDVI they are conditional on, at a **XXX**° spatial
154 scale. The rasters are freely available on GitHub (see code availability statement), along with
155 long-term (1981-2025) averages of both DENVar and the estimated mean NDVI. Since estimates of
156 variance are conditional on the corresponding estimates of the mean, we suggest users always use
157 the corresponding rasters of mean NDVI in their analyses rather than using independent estimates
158 of mean NDVI.

159 3 Results

160 Figures:

- 161 • two Robinson projections of the world: long-term mean NDVI and DENVar on spring solstice
162 OR the world with ecozone shapefiles to see if fill matches ecozone boundaries (two columns)
- 163 • hex plots of pred vs obs for mean and var (one column)
- 164 • surface plots of year (x) and latitude (y) for mean NDVI and DENVar (to detect trends over
165 space and time; two columns)
- 166 • hex plots of DENVar with other variables (add temperature, annual precip, extreme weather
167 event, fires, invasive species, max body size, disease occurrence, ...; two columns)
- 168 • supplementary material – appendix C (outputs): histograms of mean around flat map of biomes
169 (already created, but change colors?)

170 3.1 Correlations with other metrics

- 171 • temperature
- 172 • seasonal temperature range
- 173 • precip
- 174 • seasonal temperature range
- 175 • spp diversity
- 176 • gross primary productivity

- max animal weight
- <https://silvis.forest.wisc.edu/globalwui/>
- <https://silvis.forest.wisc.edu/data/dhis/>: NDVI16 (Normalized Difference Vegetation Index), EVI16 (Enhanced Vegetation Index), FPAR8 (Fraction absorbed Photosynthetically Active Radiation), LAI8 (Leaf Area Index), GPP8 (Gross Primary Productivity)

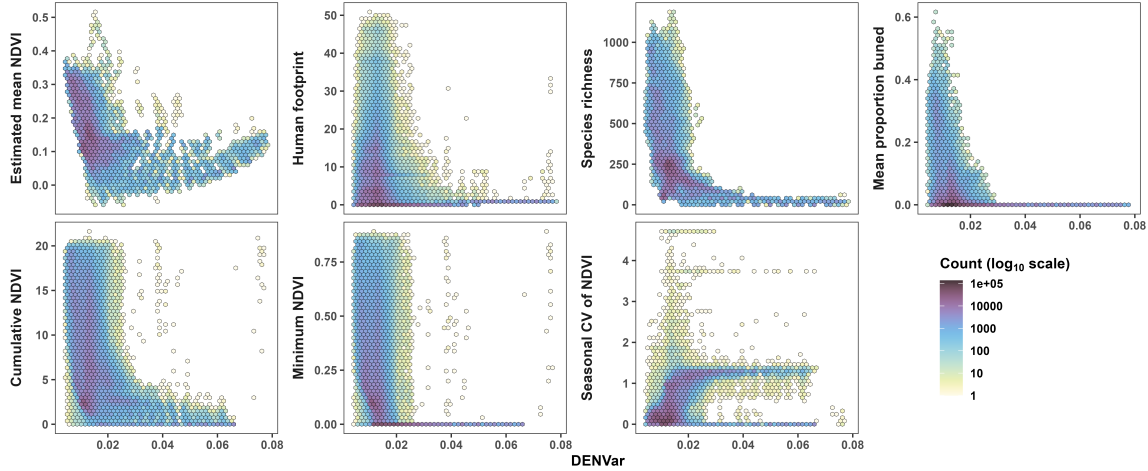


Figure 2: Hex plots of the relationship between the long-term global Dynamic Estimate of NDVI Variance (DENVar) and common ecological metrics. The fill of the hexagonal cells represents the log-10 number of points within the cell. Rowwise, the variables are: long-term mean NDVI (as estimated in this manuscript), the machine-learning Human Footprint Index (ml-HFI) of , and the 2015 cumulative, minimum, and seasonal range in the Dynamic Habitat Index of (Radeloff et al., 2019).

3.2 Obtaining the data

- Data should be available as daily rasters on a github repo?
- make an app for data annotation?

4 Discussion

We have presented a new spatiotemporally dynamic metric of environmental stochasticity, which we have called the Dynamic Estimate of NDVI Variance, DENVar. In this section, we discuss the interpretation of DENVar in greater depth, and we argue the advantages and limitations of the statistical methods used. We conclude by suggesting some use cases of DENVar within animal movement ecology, landscape ecology, and phenology, but we invite readers to consider how DENVar may be of use in their field and adjacent disciplines.

4.1 Interpreting DENVar: the importance of scale

difficulties in estimating variance

what counts as a change? predictable vs stochastic changes

smoothness of mean impacts smoothness and size of variance: depends on scale of interest (e.g., animals' ability to respond to, learn, and predict changes in the mean). Changes not attributed to trends in the mean are stochastic

DENVar is a good metric as long as the spatiotemporal scale of interest is similar or larger to that of DENVar. May not work well for small-scale processes such as the movement dynamics of a small rodent, the phenology of a small orchard when spatiotemporal heterogeneity is finer than DENVar's resolution

4.2 Environmental stochasticity and animal movement ecology

The field of animal movement ecology has long shown that animals alter their movement in response to resource abundance or availability (Burt, 1943; Southwood, 1977; Charnov, 1976; Broekman et al., 2024). More recently, movement ecologists have started focusing more on resource stochasticity (Stephens and Charnov, 1982; Rizzuto et al., 2021; Marcus and Noonan, 2023), but it is worth distinguishing between resource stochasticity and seasonality (Nilsen et al., 2005). Seasonality is generally best interpreted as a cyclical change in the mean that many animals can learn to predict, as in the case of green-wave surfing (Middleton et al., 2018; Geremia et al., 2019). In contrast, environmental stochasticity refers to unpredictable changes, such as forest fires [ref](#), sudden snow storms (Aikens et al., 2025), and unusual phenology [ref](#). Animals also respond to stochastic changes (Mezzini et al., 2025), but their ability to adapt depends on behavioral plasticity (Steixner-Kumar and Gläscher, 2020; Rickbeil et al., 2019), and memory (Polansky et al., 2015; Abrahms et al., 2019; Riotte-Lambert and Matthiopoulos, 2020; Falcón-Cortés et al., 2021; Ranc et al., 2022).

evolutionary tipping points (Botero et al., 2015)

multifarious change (Polazzo et al., 2024)

extreme events, black swans (Anderson et al., 2017; Logares and Nuñez, 2012)

- migration routes are key habitat (Ortega et al., in prep), especially in stochastic environments

- can fail to adapt to changes in conditions (Sawyer et al., 2019)

- <https://www.sciencedirect.com/science/article/pii/S0960982220308484>

- <https://onlinelibrary.wiley.com/doi/full/10.1111/gcb.15169>

- industrial development affects animals' ability to respond to both predictable and unpredictable change (<https://www.nature.com/articles/s41559-022-01887-9>)

- effects of sudden deep snow in red desert on proghorn (<https://www.sciencedirect.com/science/article/abs/pii/S0960982220308484>)

- <https://www.nature.com/articles/s41467-023-37750-z>

- <https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecy.4238>

- detecting regime shifts (in lakes: (Bjorndahl et al., 2022))

4.3 Environmental stochasticity and population dynamics

read lande_risks_1993

Environmental stochasticity reduces habitats' energetic balance and carrying capacity (Chevin et al., 2010) as individuals struggle to rely on unpredictable resources and increase the instability of populations and communities (Lande, 1993). Additionally, stochastic environments reduce the ability of specialist species to adapt and evolve while selecting for more generalist species (Levins, 1974). Consequently, stochastic environments tend to have lower species diversity (Marcus et al., 2025). Recent work by Devries has suggested that NDVI stochasticity is an important metric for evaluating forest recovery after large fires, while Collinson has suggested that variance in NDVI is an important predictor of the occurrence of lightning-induced forest fires.

• jonzen_rapid_2006

- effects can be compounded by cumulative effects (e.g., loss of species diversity; (Wolf et al., 2017))

- climate change affects plant phenology, but changes in situ may not match experimental estimates (Wolkovich et al., 2012)

- reproduction of painted lady butterfly (Stefanescu et al., 2021)

• wessling_seasonal_2018

4.4 Environmental stochasticity and fire ecology

4.5 Environmental stochasticity and phenology

5 Other useful references

- **READ**: Methods for Detecting Early Warnings of Critical Transitions in Time Series Illustrated Using Simulated Ecological Data (<https://doi.org/10.1371/journal.pone.0041010>)
- **READ**: <https://doi.org/10.1029/2024GL109298>
- **READ**: Site fidelity as a maladaptive behavior in the Anthropocene: <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecs.26000>
- **READ**: Pettorelli N., Vik J.O., Mysterud A., Gaillard J.-M., Tucker C.J. & Stenseth N.Chr. (2005). Using the satellite-derived NDVI to assess ecological responses to environmental change. *Trends in Ecology & Evolution* 20, 503–510. <https://doi.org/10.1016/j.tree.2005.05.011>
- Ecological and evolutionary consequences of changing seasonality: <https://www.science.org/doi/10.1126/science.1257500>
- increase in variance before eutrophication in lakes: <https://doi.org/10.1002/lno.10355>
- uses ndvi variance: <https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecm.1348>
- bro-jorgensen_using_2008: in the dry season preceding the rut topi antelope (*Damaliscus lunatus*) respond to forage availability at the large scale but not the fine scale
- keith_predicting_2008
- chevin_adaptation_2010
- rickbeil_plasticity_2019
- mueller_search_2008
- pettorelli_using_2005
- nilsen_can_2005
- merkle_large_2016
- tian_evaluating_2015
- huang_commentary_2021
- fan_global_2016
- wang_stochastic_2019
- pease_ecological_2024
- xu_plasticity_2021
- calibrating across sensors: <https://ieeexplore.ieee.org/abstract/document/6809983>

274 Acknowledgments

275 Anyone who made a contribution to the research or manuscript, but who is not a listed author,
276 should be acknowledged (with their permission). Types of acknowledgements include:

277 General

278 Thank others for any contributions, whether it be direct technical help or indirect assistance

279 Author Contributions

280 Describe contributions of each author to the paper, using the first initial and full last name.

281 Examples:

282 “S. Zhang conceived the idea and designed the experiments.”

283 “E. F. Mustermann and J. F. Smith conducted the experiments.”

284 “All authors contributed equally to the writing of the manuscript.”

285 Funding

286 Name financially supporting bodies (written out in full), followed by the funding awardee and asso-
287 ciated grant numbers (if applicable) in square brackets.

288 Example:

289 “This work was supported by the Engineering and Physical Sciences Research Council grant
290 numbers xxxx, yyyy; the National Science Foundation grant number zzzz; and a Leverhulme Trust
291 Research Project Grant.”

292 If the research did not receive specific funding, but was performed as part of the employment
293 of the authors, please name this employer. If the funder was involved in the manuscript writing,
294 editing, approval, or decision to publish, please declare this.

295 Conflicts of Interest

296 Conflicts of interest (COIs, also known as “competing interests”) occur when issues outside research
297 could be reasonably perceived to affect the neutrality or objectivity of the work or its assessment.

298 Authors must declare all potential interests – whether or not they actually had an influence – in a
299 ‘Conflicts of Interest’ section, which should explain why the interest may be a conflict. Authors must
300 declare current or recent funding (including for Article Processing Charges) and other payments,
301 goods or services that might influence the work. All funding, whether a conflict or not, must be
302 declared in a “Funding Statement.” The involvement of anyone other than the authors who 1) has
303 an interest in the outcome of the work; 2) is affiliated to an organization with such an interest; or 3)
304 was employed or paid by a funder, in the commissioning, conception, planning, design, conduct, or
305 analysis of the work, the preparation or editing of the manuscript, or the decision to publish must
306 be declared.

307 If there are none, the authors should state “The author(s) declare(s) that there is no conflict of
308 interest regarding the publication of this article.” Submitting authors are responsible for coauthors
309 declaring their interests. Declared conflicts of interest will be considered by the editor and reviewers
310 and included in the published article.

311 **Data Availability**

312 A data availability statement is compulsory for all research articles. This statement describes
313 whether and how others can access the data supporting the findings of the paper, including 1)
314 what the nature of the data is, 2) where the data can be accessed, and 3) any restrictions on data
315 access and why.

316 If data are in an archive, include the accession number or a placeholder for it. Also include any
317 materials that must be obtained through a Material Transfer Agreements (MTA).

318 **Supplementary Materials**

319 Describe any supplementary materials submitted with the manuscript (e.g., audio files, video clips
320 or datasets).

321 Please group supplementary materials in the following order: materials and methods, figures,
322 tables, and other files (such as movies, data, interactive images, or database files).

323 Example: Fig. S1. Title of the first supplementary figure.

324 Fig. S2. Title of the second supplementary figure.

325 Table S1. Title of the first supplementary table.

326 Data file S1. Title of the first supplementary data file.

327 Movie S1. Title of the first supplementary movie.

328 Be sure to submit all supplementary materials with the manuscript and remember to reference
329 the supplementary materials at appropriate points within the manuscript. We recommend citing
330 specific items, rather than referring to the supplementary materials in general, for example: “See
331 Figures S1-S10 in the Supplementary Material for comprehensive image analysis.”

332 A link to access the supplementary materials will be provided in the published article.

333 Supplementary Materials may include additional author notes—for example, a list of group
334 authors.

References

- Briana Abrahms, Elliott L. Hazen, Ellen O. Aikens, Matthew S. Savoca, Jeremy A. Goldbogen, Steven J. Bograd, Michael G. Jacox, Ladd M. Irvine, Daniel M. Palacios, and Bruce R. Mate. Memory and resource tracking drive blue whale migrations. *Proceedings of the National Academy of Sciences*, 116(12):5582–5587, March 2019. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.1819031116. URL <https://pnas.org/doi/full/10.1073/pnas.1819031116>.
- Ellen O. Aikens, Matthew J. Kauffman, Jerod A. Merkle, Samantha P. H. Dwinnell, Gary L. Fralick, and Kevin L. Monteith. The greenscape shapes surfing of resource waves in a large migratory herbivore. *Ecology Letters*, 20(6):741–750, June 2017. ISSN 1461-023X, 1461-0248. doi: 10.1111/ele.12772. URL <https://onlinelibrary.wiley.com/doi/10.1111/ele.12772>. Publisher: Wiley.
- Ellen O. Aikens, Jerod A. Merkle, Wenjing Xu, and Hall Sawyer. Pronghorn movements and mortality during extreme weather highlight the critical importance of connectivity. *Current Biology*, 35(8):1927–1934.e2, April 2025. ISSN 09609822. doi: 10.1016/j.cub.2025.03.010. URL <https://linkinghub.elsevier.com/retrieve/pii/S0960982225002957>.
- Sean C. Anderson, Trevor A. Branch, Andrew B. Cooper, and Nicholas K. Dulvy. Black-swan events in animal populations. *Proceedings of the National Academy of Sciences*, 114(12):3252–3257, March 2017. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.1611525114. URL <http://www.pnas.org/lookup/doi/10.1073/pnas.1611525114>.
- Hylke E. Beck, Tim R. McVicar, Albert I.J.M. Van Dijk, Jaap Schellekens, Richard A.M. De Jeu, and L. Adrian Bruijnzeel. Global evaluation of four AVHRR–NDVI data sets: Intercomparison and assessment against Landsat imagery. *Remote Sensing of Environment*, 115(10):2547–2563, October 2011. ISSN 0034-4257. doi: 10.1016/j.rse.2011.05.012. URL <https://linkinghub.elsevier.com/retrieve/pii/S0034425711001957>. Publisher: Elsevier BV.
- Muhammad Tousif Bhatti, Hammad Gilani, Muhammad Ashraf, Muhammad Shahid Iqbal, and Sarfraz Munir. Field validation of NDVI to identify crop phenological signatures. *Precision Agriculture*, 25(5):2245–2270, October 2024. ISSN 1385-2256, 1573-1618. doi: 10.1007/s11119-024-10165-6. URL <https://link.springer.com/10.1007/s11119-024-10165-6>.
- Judith A. Bjorndahl, Cale A.C. Gushulak, Stefano Mezzini, Gavin L. Simpson, Heather A. Haig, Peter R. Leavitt, and Kerri Finlay. Abrupt changes in the physical and biological structure of endorheic upland lakes due to 8-m lake-level variation during the 20th century. *Limnology and Oceanography*, 67(5):1022–1039, May 2022. ISSN 0024-3590, 1939-5590. doi: 10.1002/lno.12054. URL <https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lno.12054>.
- Tomasz Borowik, Nathalie Pettorelli, Leif Sönnichsen, and Bogumiła Jędrzejewska. Normalized difference vegetation index (NDVI) as a predictor of forage availability for ungulates in forest and field habitats. *European Journal of Wildlife Research*, 59(5):675–682, October 2013. ISSN

1612-4642, 1439-0574. doi: 10.1007/s10344-013-0720-0. URL <http://link.springer.com/10.1007/s10344-013-0720-0>.

Carlos A. Botero, Franz J. Weissing, Jonathan Wright, and Dustin R. Rubenstein. Evolutionary tipping points in the capacity to adapt to environmental change. *Proceedings of the National Academy of Sciences*, 112(1):184–189, January 2015. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.1408589111. URL <https://pnas.org/doi/full/10.1073/pnas.1408589111>.

Maarten J. E. Broekman, Jelle P. Hilbers, Selwyn Hoeks, Mark A. J. Huijbregts, Aafke M. Schipper, and Marlee A. Tucker. Environmental drivers of global variation in home range size of terrestrial and marine mammals. *Journal of Animal Ecology*, 93(4):488–500, April 2024. ISSN 0021-8790, 1365-2656. doi: 10.1111/1365-2656.14073. URL <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.14073>.

James C Bull, Nicola J Pickup, Brian Pickett, Michael P Hassell, and Michael B Bonsall. Metapopulation extinction risk is increased by environmental stochasticity and assemblage complexity. *Proceedings of the Royal Society B: Biological Sciences*, 274(1606):87–96, January 2007. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2006.3691. URL <https://royalsocietypublishing.org/doi/10.1098/rspb.2006.3691>.

William Henry Burt. Territoriality and Home Range Concepts as Applied to Mammals. *Journal of Mammalogy*, 24(3):346, August 1943. ISSN 00222372. doi: 10.2307/1374834. URL <https://academic.oup.com/jmammal/article-lookup/doi/10.2307/1374834>.

Kendall L. Calhoun, Thomas Connor, Kaitlyn M. Gaynor, Amy Van Scoyoc, Alex McInturff, Samantha E.S. Kreling, and Justin S. Brashares. Movement behavior in a dominant ungulate underlies successful adjustment to a rapidly changing landscape following megafire. *Movement Ecology*, 12(1):53, July 2024. ISSN 2051-3933. doi: 10.1186/s40462-024-00488-4. URL <https://movementecologyjournal.biomedcentral.com/articles/10.1186/s40462-024-00488-4>.

Eric L. Charnov. Optimal foraging, the marginal value theorem. *Theoretical Population Biology*, 9(2):129–136, April 1976. ISSN 00405809. doi: 10.1016/0040-5809(76)90040-X. URL <https://linkinghub.elsevier.com/retrieve/pii/004058097690040X>.

Luis-Miguel Chevin, Russell Lande, and Georgina M. Mace. Adaptation, Plasticity, and Extinction in a Changing Environment: Towards a Predictive Theory. *PLoS Biology*, 8(4):e1000357, April 2010. ISSN 1545-7885. doi: 10.1371/journal.pbio.1000357. URL <https://dx.plos.org/10.1371/journal.pbio.1000357>.

Fabio Cramer. Geodynamic diagnostics, scientific visualisation and StagLab 3.0. *Geoscientific Model Development*, 11(6):2541–2562, June 2018a. ISSN 1991-9603. doi: 10.5194/gmd-11-2541-2018. URL <https://gmd.copernicus.org/articles/11/2541/2018/>.

Fabio Cramer. Scientific colour-maps, June 2018b. URL <https://zenodo.org/record/1287763>.
Language: en.

- 407 Fabio Cramer, Grace E. Shephard, and Philip J. Heron. The misuse of colour in science
408 communication. *Nature Communications*, 11(1):5444, October 2020. ISSN 2041-1723. doi:
409 10.1038/s41467-020-19160-7. URL <https://www.nature.com/articles/s41467-020-19160-7>.
- 410 Andrea Falcón-Cortés, Denis Boyer, Evelyn Merrill, Jacqueline L. Frair, and Juan Manuel Morales.
411 Hierarchical, Memory-Based Movement Models for Translocated Elk (*Cervus canadensis*). *Frontiers in Ecology and Evolution*, 9:702925, August 2021. ISSN 2296-701X. doi: 10.3389/fevo.2021.
412 702925. URL <https://www.frontiersin.org/articles/10.3389/fevo.2021.702925/full>.
- 414 Xingwang Fan and Yuanbo Liu. A global study of NDVI difference among moderate-resolution
415 satellite sensors. *ISPRS Journal of Photogrammetry and Remote Sensing*, 121:177–191, November
416 2016. ISSN 09242716. doi: 10.1016/j.isprsjprs.2016.09.008. URL <https://linkinghub.elsevier.com/retrieve/pii/S0924271616303975>.
- 418 Christopher J. Fettig, Justin B. Runyon, Crystal S. Homicz, Patrick M. A. James, and Michael D.
419 Ulyshen. Fire and Insect Interactions in North American Forests. *Current Forestry Reports*,
420 8(4):301–316, August 2022. ISSN 2198-6436. doi: 10.1007/s40725-022-00170-1. URL <https://link.springer.com/10.1007/s40725-022-00170-1>.
- 422 Belen Franch, Eric Vermote, Jean-Claude Roger, Emilie Murphy, Inbal Becker-Reshef, Chris Justice,
423 Martin Claverie, Jyoteshwar Nagol, Ivan Csizar, Dave Meyer, Frederic Baret, Edward Masuoka,
424 Robert Wolfe, and Sadashiva Devadiga. A 30+ Year AVHRR Land Surface Reflectance Climate
425 Data Record and Its Application to Wheat Yield Monitoring. *Remote Sensing*, 9(3):296, March
426 2017. ISSN 2072-4292. doi: 10.3390/rs9030296. URL <https://www.mdpi.com/2072-4292/9/3/296>.
- 428 Richard Frankham and Barry W. Brook. The importance of time scale in conservation biology and
429 ecology. *Annales Zoologici Fennici*, 41:459–463, 2004. ISSN 0003-455X.
- 430 Michael J.M. Franklin, Richard E. Major, Michael Bedward, and Ross A. Bradstock. Relative
431 avian mobility linked to use of fire-affected resources in forested landscapes. *Forest Ecology and*
432 *Management*, 497:119484, October 2021. ISSN 03781127. doi: 10.1016/j.foreco.2021.119484. URL
433 <https://linkinghub.elsevier.com/retrieve/pii/S0378112721005739>.
- 434 Nicolas Frerebeau. *khroma: Colour Schemes for Scientific Data Visualization*. Université Bor-
435 deaux Montaigne, Pessac, France, 2024. doi: 10.5281/zenodo.1472077. URL <https://packages.tesselle.org/khroma/>.
- 437 Tak Fung, James P. O’Dwyer, and Ryan A. Chisholm. Quantifying species extinction risk under
438 temporal environmental variance. *Ecological Complexity*, 34:139–146, May 2018. ISSN 1476945X.
439 doi: 10.1016/j.ecocom.2017.09.006. URL <https://linkinghub.elsevier.com/retrieve/pii/S1476945X1730082X>.
- 441 Garmin International, Inc. and Esri. World Water Bodies, April 2025. URL <https://hub.arcgis.com/content/e750071279bf450cbd510454a80f2e63/about>.

- Chris Geremia, Jerod A. Merkle, Daniel R. Eacker, Rick L. Wallen, P. J. White, Mark Hebblewhite, and Matthew J. Kauffman. Migrating bison engineer the green wave. *Proceedings of the National Academy of Sciences*, 116(51):25707–25713, December 2019. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.1913783116. URL <http://www.pnas.org/lookup/doi/10.1073/pnas.1913783116>.
- Michael F. Goodchild and Dale A. Quattrochi. *Scale in Remote Sensing and GIS*. Routledge, New York, 1 edition, January 2023. ISBN 978-0-203-74017-0. doi: 10.1201/9780203740170. URL <https://www.taylorfrancis.com/books/9780203740170>.
- G. Hmimina, E. Dufrêne, J.-Y. Pontailier, N. Delpierre, M. Aubinet, B. Caquet, A. De Grandcourt, B. Burban, C. Flechard, A. Granier, P. Gross, B. Heinesch, B. Longdoz, C. Moureaux, J.-M. Ourcival, S. Rambal, L. Saint André, and K. Soudani. Evaluation of the potential of MODIS satellite data to predict vegetation phenology in different biomes: An investigation using ground-based NDVI measurements. *Remote Sensing of Environment*, 132:145–158, May 2013. ISSN 0034-4257. doi: 10.1016/j.rse.2013.01.010. URL <https://linkinghub.elsevier.com/retrieve/pii/S0034425713000229>. Publisher: Elsevier BV.
- Jeffrey Hollister, Tarak Shah, Jakub Nowosad, Alec L. Robitaille, Marcus W. Beck, and Mike Johnson. *elevatr: Access Elevation Data from Various APIs*. 2023. doi: 10.5281/zenodo.8335450. URL <https://github.com/jhollist/elevatr/>.
- Sha Huang, Lina Tang, Joseph P. Hupy, Yang Wang, and Guofan Shao. A commentary review on the use of normalized difference vegetation index (NDVI) in the era of popular remote sensing. *Journal of Forestry Research*, 32(1):1–6, February 2021. ISSN 1007-662X, 1993-0607. doi: 10.1007/s11676-020-01155-1. URL <https://link.springer.com/10.1007/s11676-020-01155-1>.
- Torres João, Gonçalves João, Marcos Bruno, and Honrado João. Indicator-based assessment of post-fire recovery dynamics using satellite NDVI time-series. *Ecological Indicators*, 89:199–212, June 2018. ISSN 1470-160X. doi: 10.1016/j.ecolind.2018.02.008. URL <https://linkinghub.elsevier.com/retrieve/pii/S1470160X18300852>. Publisher: Elsevier BV.
- Russell Lande. Risks of Population Extinction from Demographic and Environmental Stochasticity and Random Catastrophes. *The American Naturalist*, 142(6):911–927, 1993. ISSN 00030147, 15375323. URL <http://www.jstor.org/stable/2462690>. Publisher: [The University of Chicago Press, The American Society of Naturalists].
- Simon A. Levin. The Problem of Pattern and Scale in Ecology: The Robert H. MacArthur Award Lecture. *Ecology*, 73(6):1943–1967, December 1992. ISSN 0012-9658, 1939-9170. doi: 10.2307/1941447. URL <https://esajournals.onlinelibrary.wiley.com/doi/10.2307/1941447>.
- Richard A. Levins. *Evolution in changing environments: some theoretical explorations*. Number 2 in Monographs in population biology. Princeton Univ. Press, Princeton, NJ, 3. printing edition, 1974. ISBN 978-0-691-08062-8.
- Ramiro Logares and Martin Nuñez. Black Swans in ecology and evolution: The importance of improbable but highly influential events. *Ideas in Ecology and Evolution*, 2012. ISSN 19183178.

doi: 10.4033/iee.2012.5.5.n. URL <https://ojs.library.queensu.ca/index.php/IEE/article/view/4311>.

Paramesha Mallegowda, Ganesan Rengaiyan, Jayalakshmi Krishnan, and Madhura Niphadkar. Assessing Habitat Quality of Forest-Corridors through NDVI Analysis in Dry Tropical Forests of South India: Implications for Conservation. *Remote Sensing*, 7(2):1619–1639, February 2015. ISSN 2072-4292. doi: 10.3390/rs70201619. URL <https://www.mdpi.com/2072-4292/7/2/1619>. Publisher: MDPI AG.

Rekha Marcus and Michael J. Noonan. Changing variability is an overlooked aspect of protected area planning, October 2023. URL <http://biorxiv.org/lookup/doi/10.1101/2023.10.26.564062>.

Rekha Marcus, Stefano Mezzini, Dwija Desai, and Michael Noonan. Planning for a variable future: Environmental stochasticity, species richness, and protected areas in Canada, October 2025. URL <https://www.researchsquare.com/article/rs-7665738/v1>.

Bunkei Matsushita, Wei Yang, Jin Chen, Yuyichi Onda, and Guoyu Qiu. Sensitivity of the Enhanced Vegetation Index (EVI) and Normalized Difference Vegetation Index (NDVI) to Topographic Effects: A Case Study in High-density Cypress Forest. *Sensors*, 7(11):2636–2651, November 2007. ISSN 1424-8220. doi: 10.3390/s7112636. URL <https://www.mdpi.com/1424-8220/7/11/2636>.

Jerod A. Merkle, Kevin L. Monteith, Ellen O. Aikens, Matthew M. Hayes, Kent R. Hersey, Arthur D. Middleton, Brendan A. Oates, Hall Sawyer, Brandon M. Scurlock, and Matthew J. Kauffman. Large herbivores surf waves of green-up during spring. *Proceedings of the Royal Society B: Biological Sciences*, 283(1833):20160456, June 2016. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2016.0456. URL <https://royalsocietypublishing.org/doi/10.1098/rspb.2016.0456>.

Stefano Mezzini, Christen H. Fleming, E. Patrícia Medici, and Michael J. Noonan. How resource abundance and resource stochasticity affect organisms’ range sizes. *Movement Ecology*, 13(1):20, March 2025. ISSN 2051-3933. doi: 10.1186/s40462-025-00546-5. URL <https://movementecologyjournal.biomedcentral.com/articles/10.1186/s40462-025-00546-5>.

Arthur D. Middleton, Jerod A. Merkle, Douglas E. McWhirter, John G. Cook, Rachel C. Cook, P. J. White, and Matthew J. Kauffman. Green-wave surfing increases fat gain in a migratory ungulate. *Oikos*, 127(7):1060–1068, July 2018. ISSN 00301299. doi: 10.1111/oik.05227. URL <https://onlinelibrary.wiley.com/doi/10.1111/oik.05227>.

Pedro R. Mutti, Paulo S. Lúcio, Vincent Dubreuil, and Bergson G. Bezerra. NDVI time series stochastic models for the forecast of vegetation dynamics over desertification hotspots. *International Journal of Remote Sensing*, 41(7):2759–2788, April 2020. ISSN 0143-1161, 1366-5901. doi: 10.1080/01431161.2019.1697008. URL <https://www.tandfonline.com/doi/full/10.1080/01431161.2019.1697008>. Publisher: Informa UK Limited.

Erlend B. Nilsen, Ivar Herfindal, and John D. C. Linnell. Can intra-specific variation in carnivore home-range size be explained using remote-sensing estimates of environmental productivity? *Éco-*

516 *science*, 12(1):68–75, January 2005. ISSN 1195-6860, 2376-7626. doi: 10.2980/i1195-6860-12-1-68.
517 1. URL <https://www.tandfonline.com/doi/full/10.2980/i1195-6860-12-1-68.1>.

518 Masataka Okabe and Kei Ito. Color universal design (CUD): How to make figures and presentations
519 that are friendly to colorblind people, 2008. URL <https://jfly.uni-koeln.de/color/#pallet>.

520 David M. Olson, Eric Dinerstein, Eric D. Wikramanayake, Neil D. Burgess, George V. N. Powell,
521 Emma C. Underwood, Jennifer A. D’amico, Illanga Itoua, Holly E. Strand, John C. Morrison,
522 Colby J. Loucks, Thomas F. Allnutt, Taylor H. Ricketts, Yumiko Kura, John F. Lamoreux,
523 Wesley W. Wettengel, Prashant Hedao, and Kenneth R. Kassem. Terrestrial Ecoregions of the
524 World: A New Map of Life on Earth. *BioScience*, 51(11):933, 2001. ISSN 0006-3568. doi: 10.
525 1641/0006-3568(2001)051[0933:teotwa]2.0.co;2. URL [https://academic.oup.com/bioscience/
526 article/51/11/933-938/227116](https://academic.oup.com/bioscience/article/51/11/933-938/227116). Publisher: Oxford University Press (OUP).

527 N Pettorelli, S Ryan, T Mueller, N Bunnefeld, B Jedrzejewska, M Lima, and K Kausrud. The
528 Normalized Difference Vegetation Index (NDVI): unforeseen successes in animal ecology. *Climate
529 Research*, 46(1):15–27, January 2011. ISSN 0936-577X, 1616-1572. doi: 10.3354/cr00936. URL
530 <http://www.int-res.com/abstracts/cr/v46/n1/p15-27/>.

531 Nathalie Pettorelli, Jon Olav Vik, Atle Myrsetrud, Jean-Michel Gaillard, Compton J. Tucker,
532 and Nils Chr. Stenseth. Using the satellite-derived NDVI to assess ecological responses to en-
533 vironmental change. *Trends in Ecology & Evolution*, 20(9):503–510, September 2005. ISSN
534 01695347. doi: 10.1016/j.tree.2005.05.011. URL [https://linkinghub.elsevier.com/retrieve/
535 pii/S016953470500162X](https://linkinghub.elsevier.com/retrieve/pii/S016953470500162X).

536 Leo Polansky, Werner Kilian, and George Wittemyer. Elucidating the significance of spatial memory
537 on movement decisions by African savannah elephants using state–space models. *Proceedings of
538 the Royal Society B: Biological Sciences*, 282(1805):20143042, April 2015. ISSN 0962-8452, 1471-
539 2954. doi: 10.1098/rspb.2014.3042. URL [https://royalsocietypublishing.org/doi/10.1098/
540 rspb.2014.3042](https://royalsocietypublishing.org/doi/10.1098/rspb.2014.3042).

541 Francesco Polazzo, Romana Limberger, Frank Pennekamp, Samuel R. P.-J. Ross, Gavin L. Simpson,
542 and Owen L. Petchey. Measuring the Response Diversity of Ecological Communities Experiencing
543 Multifarious Environmental Change. *Global Change Biology*, 30(11):e17594, November 2024. ISSN
544 1354-1013, 1365-2486. doi: 10.1111/gcb.17594. URL [https://onlinelibrary.wiley.com/doi/
545 10.1111/gcb.17594](https://onlinelibrary.wiley.com/doi/10.1111/gcb.17594).

546 R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for
547 Statistical Computing, Vienna, Austria, 2024. URL <https://www.R-project.org/>.

548 V.C. Radeloff, M. Dubinin, N.C. Coops, A.M. Allen, T.M. Brooks, M.K. Clayton, G.C. Costa, C.H.
549 Graham, D.P. Helmers, A.R. Ives, D. Kolesov, A.M. Pidgeon, G. Rapacciuolo, E. Razenkova,
550 N. Suttigate, B.E. Young, L. Zhu, and M.L. Hobi. The Dynamic Habitat Indices (DHIs) from
551 MODIS and global biodiversity. *Remote Sensing of Environment*, 222:204–214, March 2019. ISSN

00344257. doi: 10.1016/j.rse.2018.12.009. URL <https://linkinghub.elsevier.com/retrieve/pii/S0034425718305625>.

Nathan Ranc, Francesca Cagnacci, and Paul R. Moorcroft. Memory drives the formation of animal home ranges: Evidence from a reintroduction. *Ecology Letters*, 25(4):716–728, April 2022. ISSN 1461-023X, 1461-0248. doi: 10.1111/ele.13869. URL <https://onlinelibrary.wiley.com/doi/10.1111/ele.13869>.

Colleen E. Reid, Laura D. Kubzansky, Jiayue Li, Jessie L. Shmool, and Jane E. Clougherty. It’s not easy assessing greenness: A comparison of NDVI datasets and neighborhood types and their associations with self-rated health in New York City. *Health & Place*, 54:92–101, November 2018. ISSN 1353-8292. doi: 10.1016/j.healthplace.2018.09.005. URL <https://linkinghub.elsevier.com/retrieve/pii/S1353829218301151>. Publisher: Elsevier BV.

Gregory J. M. Rickbeil, Jerod A. Merkle, Greg Anderson, M. Paul Atwood, Jon P. Beckmann, Eric K. Cole, Alyson B. Courtemanch, Sarah Dewey, David D. Gustine, Matthew J. Kauffman, Douglas E. McWhirter, Tony Mong, Kelly Proffitt, Patrick J. White, and Arthur D. Middleton. Plasticity in elk migration timing is a response to changing environmental conditions. *Global Change Biology*, 25(7):2368–2381, July 2019. ISSN 1354-1013, 1365-2486. doi: 10.1111/gcb.14629. URL <https://onlinelibrary.wiley.com/doi/10.1111/gcb.14629>.

Louise Riotte-Lambert and Jason Matthiopoulos. Environmental Predictability as a Cause and Consequence of Animal Movement. *Trends in Ecology & Evolution*, 35(2):163–174, February 2020. ISSN 01695347. doi: 10.1016/j.tree.2019.09.009. URL <https://linkinghub.elsevier.com/retrieve/pii/S0169534719302885>.

Matteo Rizzuto, Shawn J. Leroux, Eric Vander Wal, Isabella C. Richmond, Travis R. Heckford, Juliana Balluffi-Fry, and Yolanda F. Wiersma. Forage stoichiometry predicts the home range size of a small terrestrial herbivore. *Oecologia*, 197(2):327–338, October 2021. ISSN 0029-8549, 1432-1939. doi: 10.1007/s00442-021-04965-0. URL <https://link.springer.com/10.1007/s00442-021-04965-0>.

Hall Sawyer, Jerod A. Merkle, Arthur D. Middleton, Samantha P. H. Dwinnell, and Kevin L. Monteith. Migratory plasticity is not ubiquitous among large herbivores. *Journal of Animal Ecology*, 88(3):450–460, March 2019. ISSN 0021-8790, 1365-2656. doi: 10.1111/1365-2656.12926. URL <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.12926>.

Gavin L. Simpson. Modelling Palaeoecological Time Series Using Generalised Additive Models. *Frontiers in Ecology and Evolution*, 6:149, October 2018. ISSN 2296-701X. doi: 10.3389/fevo.2018.00149. URL <https://www.frontiersin.org/article/10.3389/fevo.2018.00149/full>.

N.T. Son, C.F. Chen, C.R. Chen, V.Q. Minh, and N.H. Trung. A comparative analysis of multitemporal MODIS EVI and NDVI data for large-scale rice yield estimation. *Agricultural and Forest Meteorology*, 197:52–64, October 2014. ISSN 0168-1923. doi: 10.1016/j.agrformet.2014.06.007.

588 URL <https://linkinghub.elsevier.com/retrieve/pii/S016819231400149X>. Publisher: El-
589 sevier BV.

590 T. R. E. Southwood. Habitat, the Templet for Ecological Strategies? *The Journal of Animal*
591 *Ecology*, 46(2):336, June 1977. ISSN 00218790. doi: 10.2307/3817. URL [https://www.jstor.](https://www.jstor.org/stable/3817?origin=crossref)
592 [org/stable/3817?origin=crossref](https://www.jstor.org/stable/3817?origin=crossref).

593 Constantî Stefanescu, Andreu Ubach, and Christer Wiklund. Timing of mating, reproductive status
594 and resource availability in relation to migration in the painted lady butterfly. *Animal Behaviour*,
595 172:145–153, February 2021. ISSN 00033472. doi: 10.1016/j.anbehav.2020.12.013. URL [https:](https://linkinghub.elsevier.com/retrieve/pii/S0003347220303742)
596 [//linkinghub.elsevier.com/retrieve/pii/S0003347220303742](https://linkinghub.elsevier.com/retrieve/pii/S0003347220303742).

597 Saurabh Steixner-Kumar and Jan Gläser. Strategies for navigating a dynamic world. *Science*, 369
598 (6507):1056–1057, August 2020. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.abd7258. URL
599 <https://www.science.org/doi/10.1126/science.abd7258>.

600 D. W. Stephens and Eric L. Charnov. Optimal foraging: Some simple stochastic models. *Behavioral*
601 *Ecology and Sociobiology*, 10(4):251–263, July 1982. ISSN 0340-5443, 1432-0762. doi: 10.1007/
602 [BF00302814](http://link.springer.com/10.1007/BF00302814). URL <http://link.springer.com/10.1007/BF00302814>.

603 Bernt-Erik Sæther. Environmental stochasticity and population dynamics of large herbivores: a
604 search for mechanisms. *Trends in Ecology & Evolution*, 12(4):143–149, April 1997. ISSN 0169-
605 5347. doi: 10.1016/S0169-5347(96)10068-9. URL [https://doi.org/10.1016/S0169-5347\(96\)](https://doi.org/10.1016/S0169-5347(96)10068-9)
606 [10068-9](https://doi.org/10.1016/S0169-5347(96)10068-9). Publisher: Elsevier.

607 United States National Aeronautics and Space Administration. Level-1 and Atmosphere Archive &
608 Distribution System Distributed Active Archive Center, 2025. URL [https://ladsweb.modaps.](https://ladsweb.modaps.eosdis.nasa.gov/archive/allData)
609 [eosdis.nasa.gov/archive/allData](https://ladsweb.modaps.eosdis.nasa.gov/archive/allData).

610 United States National Centers for Environmental Information (NCEI) and United States National
611 Aeronautics and Space Administration. NCEI Database, 2025. URL [https://ncei.noaa.gov/](https://ncei.noaa.gov/dataver)
612 [dataver](https://ncei.noaa.gov/dataver).

613 Eric Vermote and NOAA CDR Program. NOAA Climate Data Record (CDR) of AVHRR Normal-
614 ized Difference Vegetation Index (NDVI), Version 5, 2018. URL [https://www.ncei.noaa.gov/](https://www.ncei.noaa.gov/metadata/geoportal/rest/metadata/item/gov.noaa.ncdc:C01558/html)
615 [metadata/geoportal/rest/metadata/item/gov.noaa.ncdc:C01558/html](https://www.ncei.noaa.gov/metadata/geoportal/rest/metadata/item/gov.noaa.ncdc:C01558/html).

616 Eric Vermote and NOAA CDR Program. NOAA Climate Data Record (CDR) of VIIRS Normal-
617 ized Difference Vegetation Index (NDVI), Version 1, 2022. URL [https://www.ncei.noaa.gov/](https://www.ncei.noaa.gov/metadata/geoportal/rest/metadata/item/gov.noaa.ncdc:C01677/html)
618 [metadata/geoportal/rest/metadata/item/gov.noaa.ncdc:C01677/html](https://www.ncei.noaa.gov/metadata/geoportal/rest/metadata/item/gov.noaa.ncdc:C01677/html).

619 Amelia A. Wolf, Erika S. Zavaleta, and Paul C. Selman. Flowering phenology shifts in response
620 to biodiversity loss. *Proceedings of the National Academy of Sciences*, 114(13):3463–3468, March
621 2017. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.1608357114. URL [http://www.pnas.org/](http://www.pnas.org/lookup/doi/10.1073/pnas.1608357114)
622 [lookup/doi/10.1073/pnas.1608357114](http://www.pnas.org/lookup/doi/10.1073/pnas.1608357114).

- 623 E. M. Wolkovich, B. I. Cook, J. M. Allen, T. M. Crimmins, J. L. Betancourt, S. E. Travers, S. Pau,
624 J. Regetz, T. J. Davies, N. J. B. Kraft, T. R. Ault, K. Bolmgren, S. J. Mazer, G. J. McCabe, B. J.
625 McGill, C. Parmesan, N. Salamin, M. D. Schwartz, and E. E. Cleland. Warming experiments
626 underpredict plant phenological responses to climate change. *Nature*, 485(7399):494–497, May
627 2012. ISSN 0028-0836, 1476-4687. doi: 10.1038/nature11014. URL [http://www.nature.com/
628 articles/nature11014](http://www.nature.com/articles/nature11014).
- 629 Simon N. Wood. *Generalized additive models: an introduction with R*. Chapman & Hall/CRC texts
630 in statistical science. CRC Press/Taylor & Francis Group, Boca Raton, second edition edition,
631 2017. ISBN 978-1-4987-2833-1.
- 632 Simon N. Wood, Yannig Goude, and Simon Shaw. Generalized additive models for large data sets.
633 *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 64(1):139–155, January
634 2015. ISSN 00359254. doi: 10.1111/rssc.12068. URL [https://onlinelibrary.wiley.com/doi/
635 10.1111/rssc.12068](https://onlinelibrary.wiley.com/doi/10.1111/rssc.12068).
- 636 Simon N. Wood, Zheyuan Li, Gavin Shaddick, and Nicole H. Augustin. Generalized Additive
637 Models for Gigadata: Modeling the U.K. Black Smoke Network Daily Data. *Journal of the
638 American Statistical Association*, 112(519):1199–1210, July 2017. ISSN 0162-1459, 1537-274X.
639 doi: 10.1080/01621459.2016.1195744. URL [https://www.tandfonline.com/doi/full/10.1080/
640 01621459.2016.1195744](https://www.tandfonline.com/doi/full/10.1080/01621459.2016.1195744).
- 641 Xueyan Zhang, Yunfeng Hu, Dafang Zhuang, Yongqing Qi, and Xin Ma. NDVI spatial pattern
642 and its differentiation on the Mongolian Plateau. *Journal of Geographical Sciences*, 19(4):403–
643 415, August 2009. ISSN 1009-637X, 1861-9568. doi: 10.1007/s11442-009-0403-7. URL [http:
644 //link.springer.com/10.1007/s11442-009-0403-7](http://link.springer.com/10.1007/s11442-009-0403-7). Publisher: Springer Science and Business
645 Media LLC.